

# Test Plan and Results – WardWatch (Report-wiZA)

## Testing Strategy

- **Frameworks & Tools:**
  - Jest - test runner, assertions, mocking, coverage
  - React Testing Library – for component rendering & user interaction simulation
  - jest-dom – for custom DOM matchers
  - Firebase/Firestore mocks – to isolate backend logic
- **Tests are located** in src/tests/ and **run with** npm test (uses react-scripts test)
  - A single test file can be run with npm test -- src/tests/<name>.test.js.
- **CI runs tests automatically** on every push via GitHub Actions.
  - GitHub Actions (.github/workflows/push\_check.yaml) runs the full test suite automatically on every push and pull request, failing the build if any test fails.
- **Code coverage** is tracked by Codecov (evidence attached at the end of the document).
  - Codecov Coverage (automatically tracked on every push via GitHub Actions)
  - Overall coverage on main branch: 78.54% (3071 of 3910 lines covered)
  - 3-month trend: +85.98%.

## Test Cases

- Unit tests for backend services (e.g., like\_service.test.js – priority computation, add/remove like).
- Component tests for React components (e.g., resident\_dashboard.test.js – loading, error, profile, logout, cancel button).
- Integration tests for pages (e.g., public\_dashboard.test.js – data rendering, filters, empty states).
- All external dependencies (Firebase, Leaflet, React Router) are mocked.

## Test Results

- Unit tests for backend services (e.g., `like_service.test.js` – `computePriority`, `addLike`, `removeLike`).
- Component tests for React components (e.g., `resident_dashboard.test.js` – loading state, error state, profile display, logout, cancel button visibility).
- Integration tests for full pages (e.g., `public_dashboard.test.js` – data fetching, visibility settings, filters, map rendering).
- All external dependencies (Firebase Auth, Firestore, Leaflet, React Router) are mocked to ensure tests are fast and deterministic.
- **Final test run summary:**
  - **All tests passed successfully:**
  - **Test Suites: 58 passed, 58 total**
  - **Tests: 555 passed, 555 total**
  - **Execution time: ~12 seconds**

```
Test Suites: 58 passed, 58 total
Tests:      555 passed, 555 total
Snapshots:  0 total
Time:       12.045 s, estimated 16 s
Ran all test suites.
```

Figure 1: npm test run results

- **Codecov Coverage:** Screenshot of the Codecov dashboard for the main branch showing overall coverage of 78.54%, the 3-month upward trend graph, donut chart breakdown, and detailed statistics – dated 22 May 2026

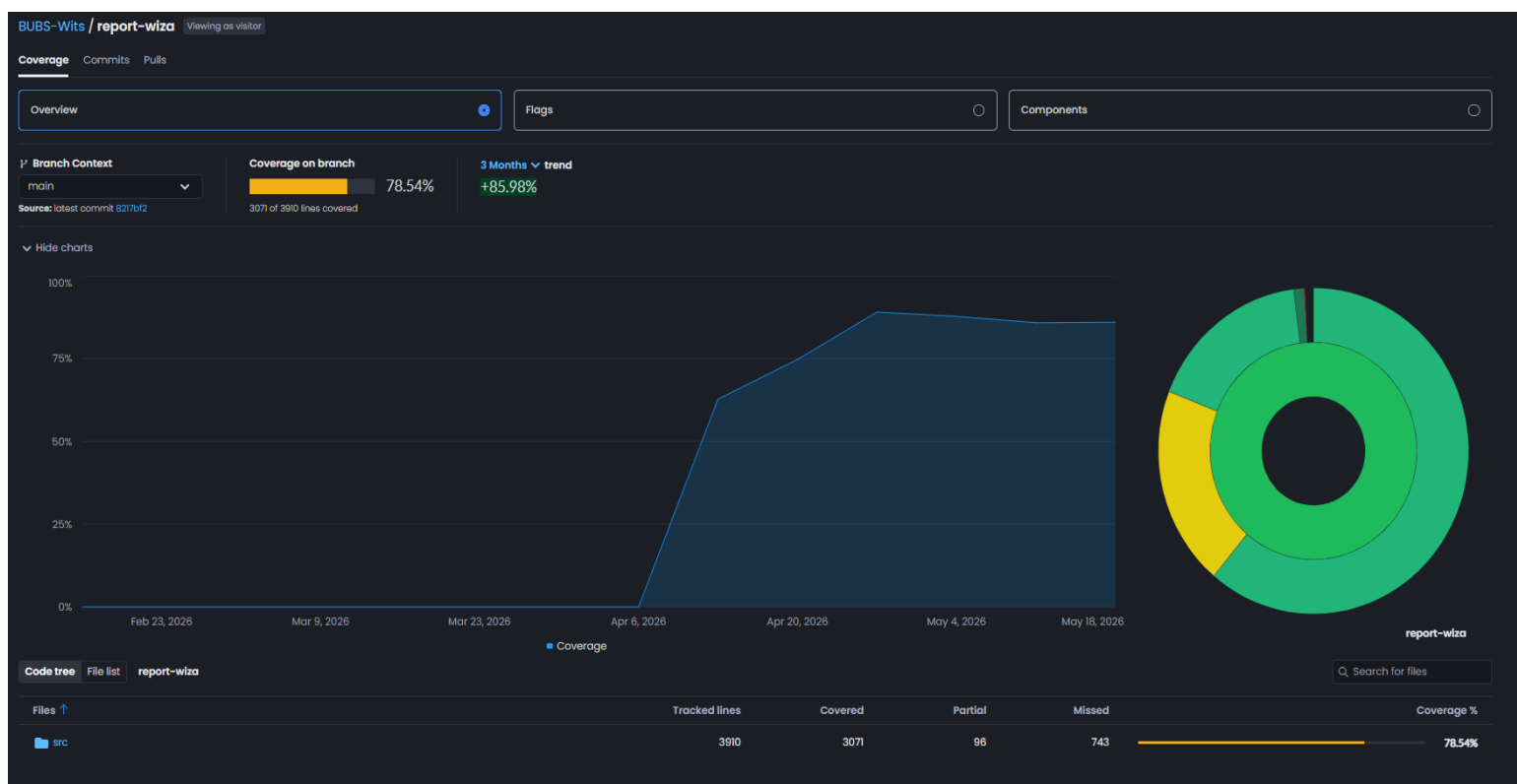


Figure 2: Codecov Coverage Dashboard